

BIO/CSC 243 Bioinformatics

Algorithm Design

Use this vocabulary list to create your algorithms. Each line must be a complete statement, all of which are denoted below by ☒. A blank _____ means that you must write something (maybe a bunch of somethings strung together) in order for it to make sense. Blanks can be filled by (1) phrases on this sheet, (2) numbers (including "a random number"), and (3) any string of characters like "AGCCTG" (or "a random string"). ### means you must replace that blank by a number. **No other words or phrases are allowed.**

Comparing two things

- ☒ • if _____
- ☒ • otherwise
- _____ is bigger than _____
- _____ is smaller than _____
- _____ is bigger than or equal to _____
- _____ is smaller than or equal to _____
- _____ is equal to _____
- _____ is not equal to _____
- and
- or
- it is not the case that _____

Repetition

- ☒ • repeat the following ### lines until a counter which starts at ### reaches ### when counting by ###
- ☒ • repeat the following ### lines as long as _____

Things you can do with storage locations (variables in memory)

- contents of storage### *(meaning whatever is held inside storage1, storage2, etc.)*
- ☒ • assign storage### to _____
- character # ### of _____ *(example: character #0 of "ATCGG")*
- total number of characters in storage###
- ☒ • return storage### *returns contents of some storage location as an answer*
- + (plus or concatenate)
- - (minus)
- * (multiply)
- / (divide)

examples

assign storage1 to random number
assign storage2 to random number

##Return sum

assign storage3 to contents of storage1 + contents of storage2
return storage3

##Return "TRUE" if the two numbers are the same. If they are not, return "FALSE"

if contents of storage1 is equal to contents of storage2
 assign storage3 to "TRUE"
 return storage3
otherwise
 assign storage3 to "FALSE"
 return storage3

##Increment number in storage1 (don't return an answer)

assign storage1 to (contents of storage1 + 1)

return the bigger number

if contents of storage1 is bigger than contents of storage2
 return storage1
otherwise
 return storage2

return "true" if thymine exists in DNA strand in storage1, else "false"

assign storage3 to 0
repeat the following 5 lines as long as contents of storage3 is smaller than total number of chars in storage1
 if character #(contents of storage3) of contents of storage1 is equal to "T"
 return "true"
 otherwise
 assign storage3 to (1+ contents of storage3)
if you're done repeating, that means you never returned "true" even though you looked at all letters
return "false"